

Solution Problem 2.3

$$(a) \quad \Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \mu = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$f(x) = \frac{1}{2\pi} \frac{1}{|\Sigma|} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

$$|\Sigma| = 1$$

$$\Sigma^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$f(x) = \frac{1}{2\pi} \cdot e^{-\frac{1}{2} \begin{bmatrix} x_1 & x_2-3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2-3 \end{bmatrix}}$$

Mahalanobis quadratic form:

$$\begin{bmatrix} x_1 & x_2-3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2-3 \end{bmatrix} =$$

$$\begin{bmatrix} x_1 & x_2-3 \end{bmatrix} \begin{bmatrix} x_1 - (x_2-3) \\ -x_1 + 2(x_2-3) \end{bmatrix} =$$

$$x_1^2 + 2(x_2-3)^2 - 2x_1(x_2-3)$$

50% confidence ellipsis:

$$x_1^2 - 2(x_2-3)^2 - 2x_1(x_2-3) \leq \chi^2_{2,0.5} = 1.39$$

$$(b) \quad x_1, \dots, x_{20} \text{ iid } \sim N_p(\mu, \Sigma) \quad n=20 \quad p=6$$

- $(x_1 - \mu)^T \Sigma^{-1} (x_1 - \mu) \sim \chi^2_6$
 - $\bar{X} \sim N_6(\mu, \Sigma/20)$
 - $\sqrt{n}(\bar{X} - \mu) \sim N_6(0, \Sigma)$
 - $n(\bar{X} - \mu)^T \Sigma^{-1} (\bar{X} - \mu) = (\bar{X} - \mu)^T \left(\frac{\Sigma}{n}\right)^{-1} (\bar{X} - \mu) \sim \chi^2_6$
 - $(n-1)S \sim W_6(\Sigma, 19)$
 - $n(\bar{X} - \mu)^T S^{-1} (\bar{X} - \mu) \sim \chi^2_6$ (approximately)
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③

$X_1, \dots, X_n \text{ iid } N_p(\mu_x, \Sigma)$

$$S_x = \frac{1}{n-1} \sum (X_i - \bar{X})(X_i - \bar{X})^T \sim \frac{1}{n-1} W_p(\Sigma, n-1)$$

$Y_1, \dots, Y_m \text{ iid } N_p(\mu_y, \Sigma)$

$$S_y = \frac{1}{m-1} \sum (Y_i - \bar{Y})(Y_i - \bar{Y})^T \sim \frac{1}{m-1} W_p(\Sigma, m-1)$$

Since Wishart distribution is reproductive in addition of degrees of freedom

$$(n-1)S_x + (m-1)S_y \sim W_p(\Sigma, n+m-2)$$

Since

$$\begin{aligned} E[(n-1)S_x + (m-1)S_y] &= E[W_p(\Sigma, n+m-2)] \\ &= (n+m-2)\Sigma \end{aligned}$$

an unbiased, pooled estimator of Σ is

$$S_p = \frac{(n-1)S_x + (m-1)S_y}{n+m-2}$$

QED.

Solution problem 2.4

Univariate uniform

$$X \sim \text{Uniform}([a, b])$$

$$\begin{aligned} E[X] &= \frac{a+b}{2} \\ \text{Var}[X] &= \frac{(a-b)^2}{12} \end{aligned}$$

Bivariate uniform with X_1, X_2 independent:

$$\begin{aligned} (X_1, X_2) &\sim 2D \text{Uniform}([a, b], [c, d]) \\ E[X] &= \begin{bmatrix} \frac{a+b}{2} \\ \frac{c+d}{2} \end{bmatrix} \end{aligned}$$

$$\Sigma = \begin{bmatrix} \frac{(a-b)^2}{12} & 0 \\ 0 & \frac{(c-d)^2}{12} \end{bmatrix}$$